

Aluno: César Eduardo de Souza

AP1Q5. Camada de transporte

Exercício sobre TCP/UDP, usando o Wireshark faça:

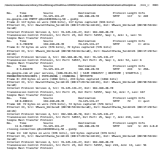
a) Analise os dump disponíveis no Moodle de REC0003 e responda para arquivo .cap as seguintes questões:

i) Qual o tipo de operação realizada?

HTTP: Solicitação de dados usando o método GET.



SMTP: Comando MAIL FROM, para envio de email.



TELNET: Operação de comando, estabelecendo uma sessão, trocando comandos e dados, e permitindo interação remota com o sistema hospedeiro.

DNS: Consulta DNS.



SSHv2: Handshake SSH.



ii) Qual é o tipo de operação realizada, dentro do escopo do protocolo?

HTTP:[Full request URI [truncated]:
<http://pagead2.googlesyndication.com/pagead/ads?client=ca-pub->

SMTP: A interação inclui comandos como "ehlo" para identificação do cliente, envio de email e comando "quit" para encerrar a sessão.

TELNET: Operação de comando, estabelecendo uma sessão, trocando comandos e dados, e permitindo interação remota com o sistema hospedeiro.

DNS: Consulta padrão (Standard Query), indicado pelos bits do cabeçalho do pacote DNS, onde o opcode é definido como "Standard query (0)", solicitando ao servidor DNS o endereço IP correspondente ao nome de domínio especificado.

SSHv2: Negociação de parâmetros de segurança, incluindo troca de chaves (Diffie-Hellman) e algoritmos de criptografia (AES 128). Inicialização da sessão.

iii) Explique a comunicação.

HTTP: Cliente com endereço IP de origem 145.254.160.237 inicia comunicação com um servidor de endereço IP 216.239.59.99, na porta 80, mediante uma solicitação HTTP GET, visando adquirir um recurso específico hospedado em "<http://pagead2.googlesyndication.com/pagead/ads>". Os cabeçalhos da solicitação contêm detalhes sobre as preferências do cliente, incluindo navegador, idiomas aceitos e codificações de conteúdo.

SMTP: Um cliente (192.168.20.70) se comunica com um servidor (74.125.131.27) para enviar um email, iniciando a comunicação com um comando "ehlo" para se identificar e iniciar a transação. Em seguida, o email é transmitido ao servidor SMTP. Após isso, o cliente encerra a sessão com o comando "quit". O servidor responde aos comandos do cliente fornecendo mensagens de boas-vindas e confirmando o encerramento da conexão.

TELNET: O cliente inicia enviando uma solicitação (SYN) para o servidor. O servidor responde com uma confirmação (SYN-ACK) e o cliente finaliza o handshake confirmando (ACK). Após a conexão ser estabelecida, cliente e servidor negociam opções, como supressão de eco, tamanho da janela, entre outros. Em seguida envia comandos ou dados para o servidor, que processa e responde conforme necessário, por exemplo, o cliente envia sequencialmente os caracteres 't', 'e', 's', 't', e o servidor responde com os mesmos caracteres. Após a conclusão da comunicação, o cliente envia um comando de encerramento da sessão, seguido pela confirmação do servidor.

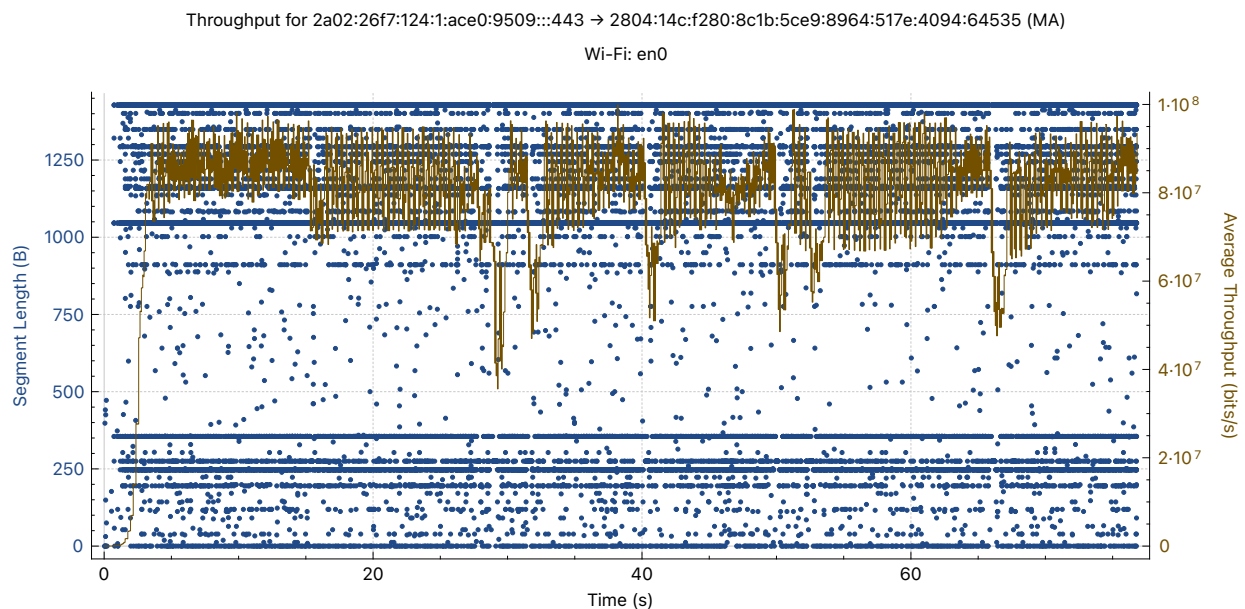
DNS: Primeiro o cliente (192.168.1.52) envia uma solicitação DNS para o servidor (8.8.8.8), contendo o nome de domínio "google.com", requisitando registros do tipo A (endereço

IP do host) na classe IN (Internet), tudo encapsulado em um pacote UDP encapsulado em um pacote IP. Então, o servidor DNS responde com uma mensagem DNS, mantendo o ID de transação para associação com a consulta original e indicando uma resposta padrão "No error". A resposta inclui uma lista de registros de recurso (RR) para "google.com", cada um com um endereço IP distinto.

SSHv2: Troca de chaves utilizando o método Diffie-Hellman. As chaves e algoritmos de criptografia, autenticação e compressão são especificados. Após a troca de chaves e a geração de uma nova chave para criptografar a comunicação, a conexão é estabelecida. Os dados são transferidos entre cliente e servidor, geralmente em várias etapas, incluindo autenticação e transferência de arquivos. O protocolo SSH utiliza criptografia AES-128 em modo de operação CBC, sem compressão.

b) Capture o tráfego TCP de um download via HTTP ou HTTPS de um conteúdo legalmente válido (deve estar respeitando os copyrights) com tamanho entre 50 e 100MB hospedado em um servidor fora do Brasil. Informar:

i) Isole em um gráfico o fluxo total e informe a URL do download.



URL:

https://rockedge.org/kernels/data/ISO/Bookworm_Pup64/BookwormPup64_10.0.6.iso

ii) Separar a abertura e fechamento da conexão de download.

Abertura:

Conversation Settings

☐ Name resolution

☐ Absolute start time

☐ Limit to display filter

Copy

Follow Stream...

Graph...

Protocol

☐ Bluetooth

☐ BPV7

☐ DCCP

☒ Ethernet

☐ FC

☐ FDDI

☐ IEEE 802.11

☐ IEEE 802.15.4

☒ IPv4

☒ IPv6

☐ IPX

☐ JXTA

☐ LTP

☐ MPTCP

☐ NCP

☐ openSAFETY

☐ RSVP

☐ SCTP

Filter list for specific type

Ethernet - 11IPv4 - 10IPv6 - 39TCP - 68UDP - 28

Address A	Port A	Address B	Port B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A
192.168.0.1	80	172.17.0.20	22	443	42	19	22	19	4	18
2603:1056:c03:1014::2	443	2804:14c:f280:8c1b:5ce9:8964:517e:4094	63031	31	13	16	18	8	3	18
2603:1063:204::1f	443	2804:14c:f280:8c1b:5ce9:8964:517e:4094	63370	14	1	12	7	749	bytes	12
2620:149:a41:580::2:3	443	2804:14c:f280:8c1b:5ce9:8964:517e:4094	64736	16	1	3	10	875	bytes	16
2620:149:a41:580::4:7	443	2804:14c:f280:8c1b:5ce9:8964:517e:4094	64737	9	904	bytes	4	4	409	bytes
2620:1ec:8f8::10	443	2804:14c:f280:8c1b:5ce9:8964:517e:4094	64746	8	6	0	4	6	0	4
2620:1ec:8fa::10	443	2804:14c:f280:8c1b:5ce9:8964:517e:4094	64743	2	2	9	1	2	0	2
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64769	2600:1419:2a00:285::2903	443	38	13	34	19	3	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64770	2600:1419:2a00:285::2903	443	36	13	35	18	3	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64766	2600:1419:2a00:288::2903	443	36	13	31	18	3	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64775	2600:1419:2a00:4::5f65:1993	80	10	1	41	6	841	bytes	31
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64772	2600:1419:4e00:e::58dd:66	443	28	16	37	12	3	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64765	2600:1419:4e00:e::58dd:78	443	25	7	30	11	2	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64768	2600:1419:4e00:e::58dd:78	443	24	7	33	10	2	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64773	2600:1419:4e00:e::58dd:78	443	27	8	38	12	2	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	59762	2603:1030:210::f:1	443	15	2	20	9	798	bytes	18
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64757	2603:1056:2000:30::12	443	34	15	18	18	5	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64783	2603:1056:c01:1002::2	443	39	16	51	16	6	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64784	2603:1056:c01:113e::2	443	51	20	52	20	9	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	63032	2603:1056:c03:1014::2	443	468	192	17	147	118	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64780	2603:1056:c03:2096::2	443	31	12	48	13	5	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64758	2603:1057:2:38::2	443	38	24	19	17	3	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64782	2620:149:a41:580::2:1	443	3	266	bytes	50	2	172	bytes
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64785	2620:149:a41:580::2:1	443	31	13	53	15	4	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64786	2620:149:a41:580::2:1	443	31	13	54	14	4	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64748	2620:149:a41:580::2:3	443	93	35	1	46	18	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64789	2620:149:a41:580::2:3	443	41	13	59	18	4	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64790	2620:149:a41:580::2:3	443	176	139	61	73	10	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64764	2620:149:a41:580::2:7	443	17	8	29	9	1	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64763	2620:149:a41:580::2:8	443	32	15	28	16	5	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64752	2620:149:a41:580::2:9	443	31	15	7	14	3	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64781	2620:149:a41:580::2:b	443	27	12	49	12	3	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64754	2620:1ec:8f8::10	443	22	9	11	9	1	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64756	2620:1ec:8f8::10	443	27	17	15	13	2	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64760	2620:1ec:8f8::10	443	13	8	22	5	911	bytes	22
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64762	2620:1ec:8f8::10	443	35	20	24	19	2	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64771	2620:1ec:8f8::10	443	32	20	36	16	2	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64777	2620:1ec:8f8::10	443	30	20	45	14	2	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64779	2620:1ec:8f8::10	443	34	20	47	18	2	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64788	2620:1ec:8f8::10	443	14	8	58	6	985	bytes	58
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64792	2620:1ec:8f8::10	443	28	18	63	13	2	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64794	2620:1ec:8f8::10	443	28	18	65	13	2	0	3
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64749	2620:1ec:8fa::10	443	40	13	2	18	4	0	3

Help

Close

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Address A	Port A	Address B	Port B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64783	2603:1056:c01:1002::2	443	39	16	51	16	6	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64784	2603:1056:c01:113e::2	443	51	20	52	20	9	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	63032	2603:1056:c03:1014::2	443	468	192	17	147	118	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64780	2603:1056:c03:2096::2	443	31	12	48	13	5	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64758	2603:1057:2:38::2	443	38	24	19	17	3	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64782	2620:149:a41:580::21	443	3	266	50	2	172	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64785	2620:149:a41:580::21	443	31	13	53	15	4	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64786	2620:149:a41:580::21	443	31	13	54	14	4	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64748	2620:149:a41:580::23	443	93	35	61	1	46	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64789	2620:149:a41:580::23	443	41	13	59	18	4	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64790	2620:149:a41:580::23	443	176	139	61	73	10	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64764	2620:149:a41:580::27	443	17	8	29	9	1	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64763	2620:149:a41:580::28	443	32	15	28	16	5	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64752	2620:149:a41:580::28	443	31	15	28	7	14	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64781	2620:149:a41:580::2b	443	27	12	49	12	3	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64754	2620:1ec:8f8::10	443	22	9	11	9	1	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64756	2620:1ec:8f8::10	443	27	17	15	13	2	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64760	2620:1ec:8f8::10	443	13	8	22	5	911	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64762	2620:1ec:8f8::10	443	35	20	24	19	2	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64771	2620:1ec:8f8::10	443	32	20	36	16	2	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64777	2620:1ec:8f8::10	443	30	20	45	14	2	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64779	2620:1ec:8f8::10	443	34	20	47	18	2	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64788	2620:1ec:8f8::10	443	14	8	58	6	985	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64792	2620:1ec:8f8::10	443	28	18	63	13	2	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64794	2620:1ec:8f8::10	443	28	18	65	13	2	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64749	2620:1ec:8f8::10	443	40	13	2	18	4	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64753	2620:1ec:8f8::10	443	41	14	10	19	4	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64755	2620:1ec:8f8::10	443	38	13	13	17	4	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64759	2620:1ec:8f8::10	443	38	13	13	17	4	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64761	2620:1ec:8f8::10	443	42	14	23	21	5	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64767	2620:1ec:8f8::10	443	40	13	32	19	4	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64774	2620:1ec:8f8::10	443	41	14	39	18	4	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64776	2620:1ec:8f8::10	443	42	14	44	19	4	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64778	2620:1ec:8f8::10	443	40	13	46	18	4	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64787	2620:1ec:8f8::10	443	37	13	56	18	4	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64791	2620:1ec:8f8::10	443	40	13	62	19	4	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64793	2620:1ec:8f8::10	443	40	14	64	19	5	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64795	2620:1ec:8f8::10	443	38	13	66	18	4	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64796	2620:1ec:8f8::10	443	39	13	67	17	4	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64750	2a02:26f7:124:3ace0:9509::	443	12	2	5	9	2	0	0
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64751	2a02:26f7:124:3ace0:9585::	443	766	271	887	16	32	0	0
fe80:1066:17bf:97de:5799	49152	fe80:1ca0:bc40:e256:3422	59980	10	3	14	5	2	0	0
fe80:1ca0:bc40:e256:3422	59980	fe80:1066:17bf:97de:5799	49152	68	3	8	22	7	0	0

Help

Close

iii) Identificar eventuais perdas de pacotes, retransmissões e resets (RST). Definir alguma hipótese fundamentada em dados da coleta para o comportamento da transferência?

The screenshot shows a Wireshark capture of a TCP connection. The filter bar at the top is set to `(tcp.analysis.lost_segment || tcp.analysis.retransmission || tcp.flags.reset == 1)`. The packet list on the left shows a series of packets, with some marked as retransmissions (e.g., packet 443). The packet details pane on the right shows the structure of a TCP segment, including the header and payload. The packet bytes pane at the bottom shows the raw data of the selected packet.

Address A	Port A	Address B	Port B	Packets	Bytes	Stream ID	Total Packets	Percent Filtered
2603:1056:c03:1014::2	443	2804:14c:f280:8c1b:5ce9:8964:517e:4094	63031	2	3 kB	16	31	6.45%
2620:149:a41:580::2:3	443	2804:14c:f280:8c1b:5ce9:8964:517e:4094	64736	8	640 bytes	3	16	50.00%
2620:149:a41:580::4:7	443	2804:14c:f280:8c1b:5ce9:8964:517e:4094	64737	3	234 bytes	4	9	33.33%
2620:1ec:8f8::10	443	2804:14c:f280:8c1b:5ce9:8964:517e:4094	64746	7	5 kB	0	8	87.50%
2620:1ec:8fa::10	443	2804:14c:f280:8c1b:5ce9:8964:517e:4094	64743	1	74 bytes	9	2	50.00%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64769	2600:1419:2a00:285::2903	443	5	382 bytes	34	38	13.16%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64770	2600:1419:2a00:285::2903	443	3	222 bytes	35	36	8.33%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64766	2600:1419:2a00:288::2903	443	3	222 bytes	31	36	8.33%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64772	2600:1419:4e00:e:58dd:66	443	1	74 bytes	37	28	3.57%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64765	2600:1419:4e00:e:58dd:78	443	1	86 bytes	30	25	4.00%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64768	2600:1419:4e00:e:58dd:78	443	1	86 bytes	33	24	4.17%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64773	2600:1419:4e00:e:58dd:78	443	3	222 bytes	38	27	11.11%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64783	2603:1056:c01:1002::2	443	1	74 bytes	51	39	2.56%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64784	2603:1056:c01:113e::2	443	1	74 bytes	52	51	1.96%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	63032	2603:1056:c03:1014::2	443	5	8 kB	17	468	1.07%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64780	2603:1056:c03:2096::2	443	1	74 bytes	48	31	3.23%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64782	2620:149:a41:580::2:1	443	1	74 bytes	50	3	33.33%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64785	2620:149:a41:580::2:1	443	1	86 bytes	53	31	3.23%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64786	2620:149:a41:580::2:1	443	1	86 bytes	54	31	3.23%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64748	2620:149:a41:580::2:3	443	2	192 bytes	1	93	2.15%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64789	2620:149:a41:580::2:3	443	1	98 bytes	59	41	2.44%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64790	2620:149:a41:580::2:3	443	5	426 bytes	61	176	2.84%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64764	2620:149:a41:580::2:7	443	5	370 bytes	29	17	29.41%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64781	2620:149:a41:580::2:b	443	1	86 bytes	49	27	3.70%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64754	2620:1ec:8f8::10	443	4	296 bytes	11	22	18.18%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64756	2620:1ec:8f8::10	443	15	10 kB	15	27	55.56%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64762	2620:1ec:8f8::10	443	23	13 kB	24	35	65.71%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64771	2620:1ec:8f8::10	443	20	13 kB	36	32	62.50%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64777	2620:1ec:8f8::10	443	18	12 kB	45	30	60.00%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64779	2620:1ec:8f8::10	443	22	13 kB	47	34	64.71%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64788	2620:1ec:8f8::10	443	2	148 bytes	58	14	14.29%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64792	2620:1ec:8f8::10	443	16	11 kB	63	28	57.14%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64794	2620:1ec:8f8::10	443	16	11 kB	65	28	57.14%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64749	2620:1ec:8f8::10	443	1	98 bytes	2	40	2.50%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64753	2620:1ec:8f8::10	443	2	196 bytes	10	41	4.88%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64755	2620:1ec:8f8::10	443	1	98 bytes	13	38	2.63%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64759	2620:1ec:8f8::10	443	2	196 bytes	21	38	5.26%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	6476	2620:1ec:8f8::10	443	4	885 bytes	23	42	9.52%

The screenshot shows a Wireshark capture of a TCP connection. The filter bar at the top is set to `(tcp.analysis.lost_segment || tcp.analysis.retransmission || tcp.flags.reset == 1)`. The packet list on the left shows a series of packets, with some marked as retransmissions (e.g., packet 443). The packet details pane on the right shows the structure of a TCP segment, including the header and payload. The packet bytes pane at the bottom shows the raw data of the selected packet.

Address A	Port A	Address B	Port B	Packets	Bytes	Stream ID	Total Packets	Percent Filtered
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64773	2600:1419:4e00:e:58dd:78	443	3	222 bytes	38	27	11.11%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64783	2603:1056:c01:1002::2	443	1	74 bytes	51	39	2.56%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64784	2603:1056:c01:113e::2	443	1	74 bytes	52	51	1.96%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	63032	2603:1056:c03:1014::2	443	5	8 kB	17	468	1.07%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64780	2603:1056:c03:2096::2	443	1	74 bytes	48	31	3.23%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64782	2620:149:a41:580::2:1	443	1	74 bytes	50	3	33.33%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64785	2620:149:a41:580::2:1	443	1	86 bytes	53	31	3.23%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64786	2620:149:a41:580::2:1	443	1	86 bytes	54	31	3.23%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64748	2620:149:a41:580::2:3	443	2	192 bytes	1	93	2.15%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64789	2620:149:a41:580::2:3	443	1	98 bytes	59	41	2.44%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64790	2620:149:a41:580::2:3	443	5	426 bytes	61	176	2.84%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64764	2620:149:a41:580::2:7	443	5	370 bytes	29	17	29.41%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64781	2620:149:a41:580::2:b	443	1	86 bytes	49	27	3.70%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64754	2620:1ec:8f8::10	443	4	296 bytes	11	22	18.18%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64756	2620:1ec:8f8::10	443	15	10 kB	15	27	55.56%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64762	2620:1ec:8f8::10	443	23	13 kB	24	35	65.71%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64771	2620:1ec:8f8::10	443	20	13 kB	36	32	62.50%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64777	2620:1ec:8f8::10	443	18	12 kB	45	30	60.00%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64779	2620:1ec:8f8::10	443	22	13 kB	47	34	64.71%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64788	2620:1ec:8f8::10	443	2	148 bytes	58	14	14.29%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64792	2620:1ec:8f8::10	443	16	11 kB	63	28	57.14%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64794	2620:1ec:8f8::10	443	16	11 kB	65	28	57.14%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64749	2620:1ec:8f8::10	443	1	98 bytes	2	40	2.50%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64753	2620:1ec:8f8::10	443	2	196 bytes	10	41	4.88%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64755	2620:1ec:8f8::10	443	1	98 bytes	13	38	2.63%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64759	2620:1ec:8f8::10	443	2	196 bytes	21	38	5.26%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64761	2620:1ec:8f8::10	443	4	885 bytes	23	42	9.52%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64774	2620:1ec:8f8::10	443	1	98 bytes	39	41	2.44%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64776	2620:1ec:8f8::10	443	1	98 bytes	44	42	2.38%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64778	2620:1ec:8f8::10	443	1	98 bytes	46	40	2.50%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64787	2620:1ec:8f8::10	443	1	98 bytes	56	37	2.70%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64791	2620:1ec:8f8::10	443	1	98 bytes	62	40	2.50%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64793	2620:1ec:8f8::10	443	2	689 bytes	64	40	5.00%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64750	2a02:26f7:124:1ace0:9509::	443	6	1 kB	5	12	50.00%
2804:14c:f280:8c1b:5ce9:8964:517e:4094	64751	2a02:26f7:124:3ace0:9585::	443	2	3 kB	6	766	2.21%
fe80::1066:17bf:97de:979	49152	fe80::1ca0:bca0:e256:3422	59980	1	86 bytes	14	10	10.00%
fe80::1ca0:bca0:e256:3422	62772	fe80::1066:17bf:97de:979	57241	2	148 bytes	40	174	1.15%
fe80::1ca0:bca0:e256:3422	62773	fe80::1ca0:bca0:e256:3422	57243	1	74 bytes	42	162	0.62%

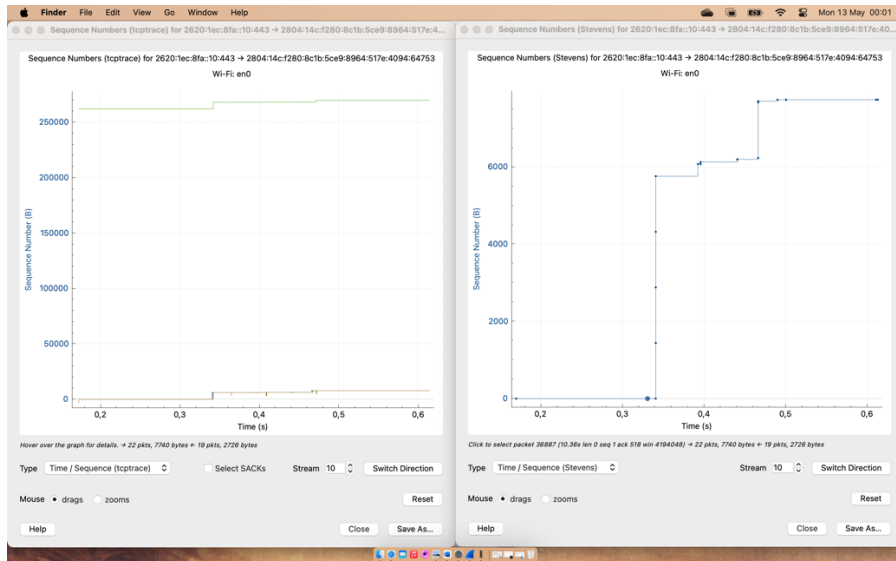
iv) Indicar o tamanho máximo das janelas (rwnd e cwnd) durante a conexão.

The image shows a Wireshark packet capture of a TCP connection. The filter is '(tcp.flags.syn == 1 & tcp.flags.ack == 0) && ip.v6.addr == 2804:14c:f280:8c1b:5ce9:8964:517e:4094'. The table below represents the data visible in the packet list pane.

No.	Time	Source	Destination	Protocol	Length	Info
1348	1.342438	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	64748 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=308484178 TSecr=0
5773	1.780386	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	64749 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1564208018 TSecr=0
10884	1.778291	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	[TCP Retransmission] 64748 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64
10885	1.778291	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:14c:a4:1007:2:3	TCP	98	[TCP Retransmission] 64748 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64
21495	7.932494	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	64750 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=425093310 TSecr=0
20740	7.838096	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:207f:1241:1ac6b:55091	TCP	98	[TCP Retransmission] 64750 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64
20741	7.837534	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:207f:1241:1ac6b:55091	TCP	98	64751 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=307933120 TSecr=0
20742	7.838586	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:207f:1241:1ac6b:55091	TCP	98	[TCP Retransmission] 64750 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64
20692	8.51237	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:14c:a4:1500:2:9	TCP	98	64752 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=254238733 TSecr=0
33181	16.420763	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	64753 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1283232893 TSecr=0
33147	16.387515	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	[TCP Retransmission] 64753 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64
35482	16.329534	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	64754 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1314610645 TSecr=0
33658	16.377877	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	[TCP Retransmission] 64753 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64
342995	16.511978	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	64755 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=3079088833 TSecr=0
144772	18.507880	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	[TCP Retransmission] 64755 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64
144749	18.453902	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	64756 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1337709480 TSecr=0
371334	20.512747	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	64757 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=3208081934 TSecr=0
182412	21.338812	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:1807:2:38c:12	TCP	98	64758 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1208086971 TSecr=0
254826	25.080388	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	64759 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1307877217 TSecr=0
250141	26.974243	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	[TCP Retransmission] 64759 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64
250133	27.004488	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	64760 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1257373310 TSecr=0
250129	27.004084	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	[TCP Retransmission] 64759 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64
362429	35.462429	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	64761 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1602121401 TSecr=0
361429	35.511429	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	[TCP Retransmission] 64761 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64
361422	35.505153	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	64762 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1304037108 TSecr=0
367008	35.687154	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	[TCP Retransmission] 64761 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64
366991	35.672549	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	64763 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1854368926 TSecr=0
367522	35.766808	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	[TCP Retransmission] 64762 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64
480433	38.455588	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:14c:a4:1500:2:8	TCP	98	64763 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1854368926 TSecr=0
480498	38.715511	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:14c:a4:1500:2:7	TCP	98	64764 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1506665064 TSecr=0
471541	44.899922	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:14c:a4:1500:2:1	TCP	98	64765 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=3082280859 TSecr=0
471542	44.891198	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:14c:a4:1500:2:1	TCP	98	64766 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1458059594 TSecr=0
472124	44.177824	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	64767 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1304047225 TSecr=0
472481	44.206196	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:14c:a4:1500:2:1	TCP	98	64768 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1308422189 TSecr=0
472434	44.200207	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:14c:a4:1500:2:1	TCP	98	64769 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1304278156 TSecr=0
472780	44.270511	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:14c:a4:1500:2:1	TCP	98	64770 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1312180824 TSecr=0
472738	44.265559	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	64771 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1373052415 TSecr=0
472778	44.293971	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:14c:a4:1500:2:1	TCP	98	64772 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1254940884 TSecr=0
473131	44.397523	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:14c:a4:1500:2:1	TCP	98	64773 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1484145474 TSecr=0
377648	52.626156	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	64774 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=2857781515 TSecr=0
377649	52.626156	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	[TCP Retransmission] 64774 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64
421522	56.355758	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:14c:a4:1500:2:1	TCP	98	64775 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1845515398 TSecr=0
888974	61.872332	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	64776 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=27815087 TSecr=0
888975	61.872332	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	[TCP Retransmission] 64776 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64
888976	61.872332	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	64777 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1487204116 TSecr=0
787878	69.578684	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	64778 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1371534935 TSecr=0
788568	69.645422	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	[TCP Retransmission] 64778 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64
788569	69.636633	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	64779 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1374138818 TSecr=0
838632	73.218674	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:1806:03:1206:2	TCP	98	64780 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1228737375 TSecr=0
832080	73.212659	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:14c:a4:1500:2:1	TCP	98	64781 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1276230995 TSecr=0
834955	73.578051	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:14c:a4:1500:2:1	TCP	98	64782 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=154294263 TSecr=0
839556	73.934478	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:1806:03:1206:2	TCP	98	64783 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=128287882 TSecr=0
841242	74.526169	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:1806:03:123c:12	TCP	98	64784 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=128287882 TSecr=0
848996	74.557125	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:14c:a4:1500:2:1	TCP	98	64785 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1397782959 TSecr=0
851888	75.630784	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:14c:a4:1500:2:1	TCP	98	64786 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1208542395 TSecr=0
896483	76.363879	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	64787 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64 TSval=1371537138 TSecr=0
897206	76.524888	2080:14c:f280:8c1b:5ce9:8964:517e:4094	2020:10c:8fa:10:443	TCP	98	[TCP Retransmission] 64787 → 443 [SYN] Seq=65535 Len=0 MSS=1440 WS=64

65535

v) Analise a relação entre número de sequência e tempo usando time-sequence graph (stevens e tcptrace).



Relação: Enquanto o tempo passa, o número de sequência aumenta.

